

scenario	what the slide verifies
turning	with the gain selected from root locus, the vessel can complete the heading change, but the path still shows transient lag
obstacle avoidance	the designed gain lets the ship bypass an obstacle smoothly, showing that the root-locus choice is not limited to a step input
one-minute turning with $K = 0.6064$	the heading response follows the command, while the trajectory reveals the practical turning-circle bias of the designed controller
one-minute turning with $K = 20$	a much larger gain changes the maneuvering trace noticeably, so gain selection must be judged in task space, not only on the pole map